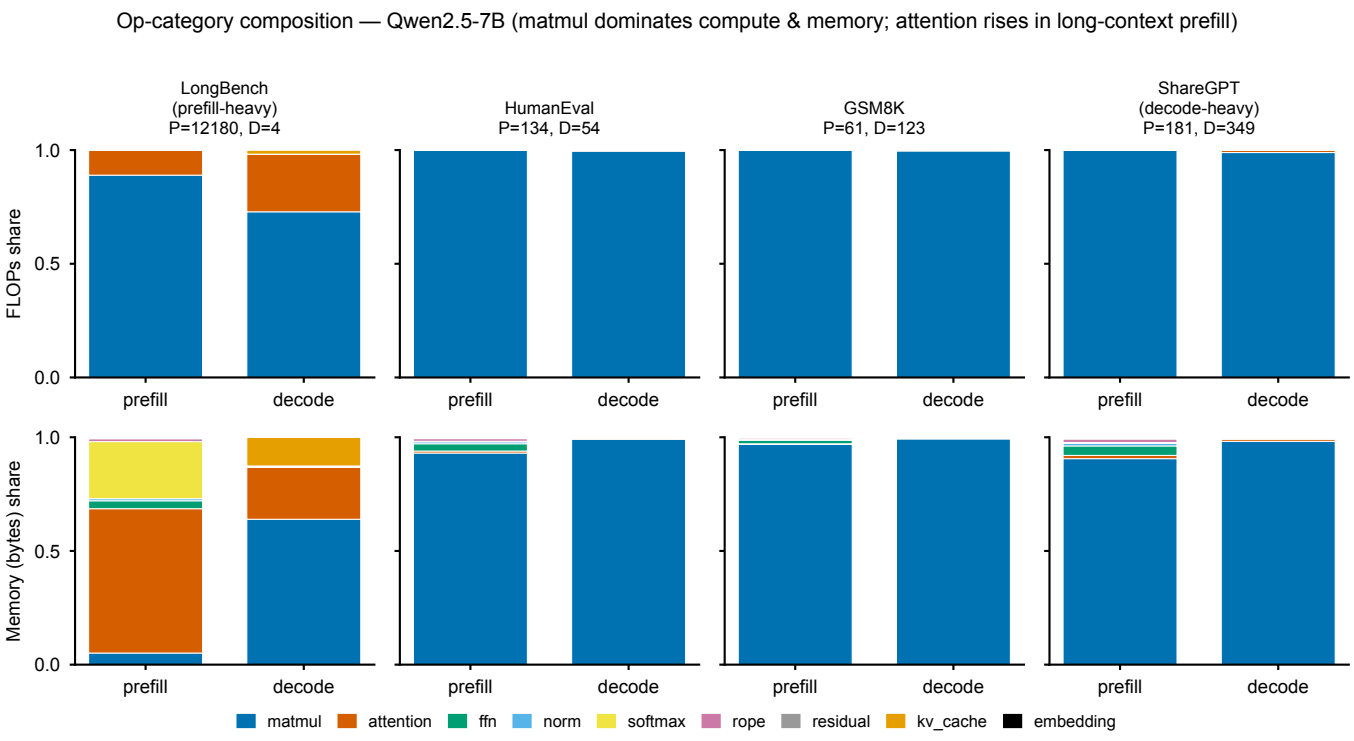
Op-category composition — Qwen2.5-7B (matmul dominates compute & memory; attention rises in long-context prefill)
LongBench
(prefill-heavy)
P=12180, D=4
HumanEval
P=134, D=54
GSM8K
P=61, D=123
ShareGPT
(decode-heavy)
P=181, D=349
FLOPs share
Memory (bytes) share
1.0
0.5
0.0
prefill
decode
matmul
attention
ffn
norm
softmax
rope
residual
kv_cache
embedding